ALICE L. WALTON SCHOOL OF MEDICINE

Metformin, GLP-1 RA, and SGLT2 Inhibitors

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Explain the mechanism of action of metformin, GLP-1 receptor agonists, and SGLT2 inhibitors in the context of managing Type 2 Diabetes.
2. Discuss the adverse effects of metformin, GLP-1 receptor agonists, and SGLT2 inhibitors and discuss strategies to minimize/prevent such effects.
3. Describe the potential risk of developing lactic acidosis based on metformin pharmacokinetics.

CASE CONNECTION

TK is a 48-year-old male recently diagnosed with Type 2 diabetes mellitus during a routine check-up. You saw him on one of your ECE clinic days and decided to initiate lifestyle modifications along with metformin. He returns to the clinic 10 days later for follow up labs, which show some improvement. He endorses persistent diarrhea, bloating, and occasional nausea. He denies fever, blood in stool, or recent dietary changes. Despite the discomfort, he has been adherent to his medication.

TK is not sure he can "take much more of all these bathroom trips" and is wondering if he should stop taking it altogether.

As you are talking with TK, are there any tips/adjustments you can make to help him? What strategies can help reduce TK's symptoms while maintaining metformin therapy?

GO TO CONCLUSION ↓

What is Type 2 Diabetes Mellitus?

Type 2 Diabetes Mellitus (DM) will be discussed extensively in the endocrine course. Briefly, it is initially characterized by insulin resistance, followed by high glucagon levels and insufficient insulin secretion. Obesity, particularly excess intra-abdomin h a sedentary lifestyle, poor diet, smoking, alcohol consumption, and sleep deprivation. Genetic factors also play a significant role, especially if a parent has Type 2 DM.

Several medication classes can be used to manage Type 2 DM. This brick focuses on three commonly used non-insulin medication classes: metformin, glucagon-like peptide-1 receptor agonists (GLP-1 RAs), and sodium-glucose cotransporter 2 inhibitors (SGLT2 inhibitors). These are not the only medications used for Type 2 DM, but they are especially important because they lower glucose with a low risk of hypoglycemia when used without insulin or insulin secretagogues. In addition, selected GLP-1 RAs and SGLT2 inhibitors have become especially common because they can provide benefits beyond glucose lowering, including cardiovascular risk reduction and kidney protection in appropriate patients. Current diabetes standards emphasize using GLP-1 RAs and/or SGLT2 inhibitors with proven benefit in patients with heart failure, chronic kidney disease, established cardiovascular disease, or multiple cardiovascular risk factors, independent of background metformin use or A1c.

Insulin resistance is caused by elevated free fatty acids, reduced insulin receptors, cytokines from adipocytes, leptin deficiency, and high cortisol levels. Inadequate pancreatic insulin secretion further exacerbates the condition, leading to ***hyperglycemia*** when the pancreas cannot produce enough insulin to compensate for the resistance. These concepts are shown in Figure 1. This concept is important therapeutically because Type 2 DM can be targeted through several mechanisms: improving insulin sensitivity, enhancing glucose-dependent insulin secretion, decreasing glucagon signaling, slowing gastric emptying, or increasing urinary glucose excretion.

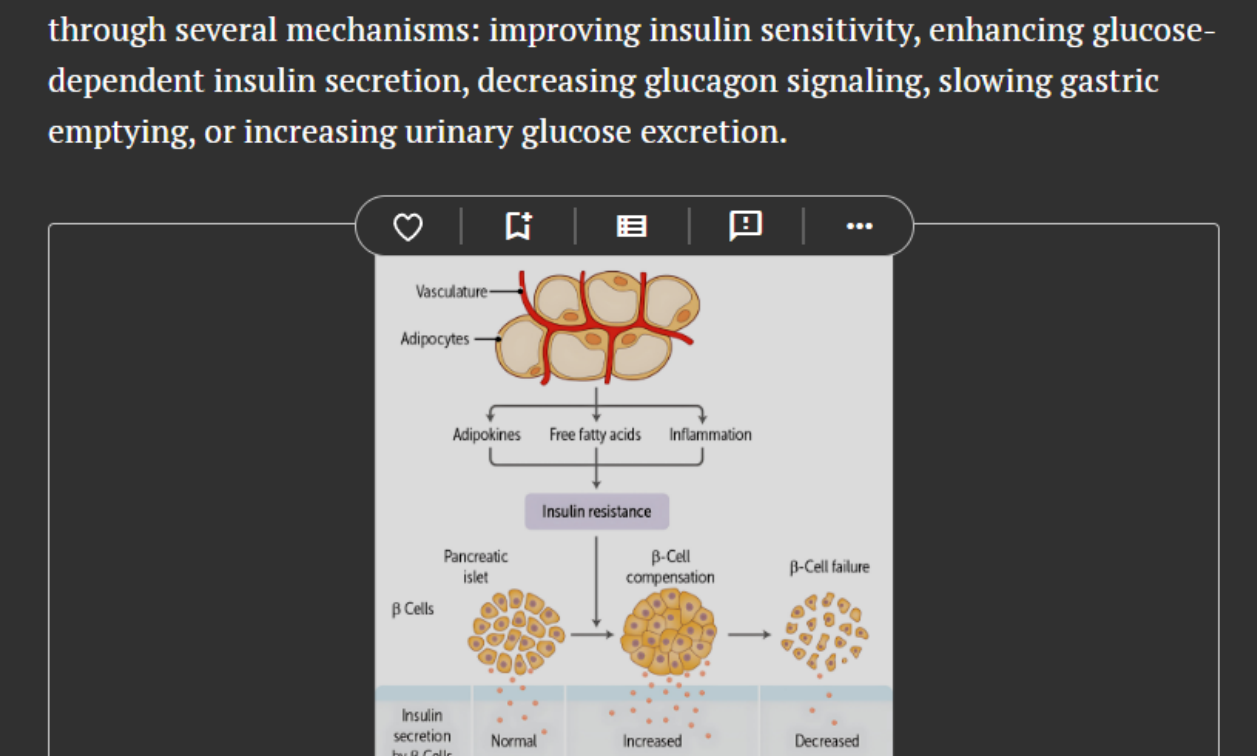

Figure 1

How do metformin, GLP-1 receptor agonists, and SGLT2 inhibitors work? (LO#1)

Table 1 provides and overview of the medications we will discuss in this brick.

Table 1

Mechanistic category	Medication class	Example drugs	Primary mechanism
Drugs that increase cellular sensitivity to insulin	Biguanides	Metformin	Decreases hepatic glucose production and improves peripheral insulin sensitivity
Drugs that stimulate glucose-dependent insulin release	GLP-1 receptor agonists	Semaglutide, liraglutide, dulaglutide, exenatide	Mimic incretin signaling to increase glucose-dependent insulin secretion, decrease glucagon secretion, slow gastric emptying, and increase satiety
Drugs that increase renal excretion of glucose	SGLT2 inhibitors	Empagliflozin, dapagliflozin, canagliflozin	Block glucose reabsorption in the proximal tubule, increasing urinary glucose excretion

Metformin

Metformin is in the biguanide class, and it is usually one of, if not the, first drugs used to treat Type 2 DM. It can also be used for patients with prediabetes. Its ultimate effect is to *increase cellular sensitivity to insulin.*

Metformin exerts its action primarily in the liver, skeletal muscle, and intestine (Figure 2). It decreases hepatic glucose production (decreased gluconeogenesis and glycogenolysis), decreases intestinal absorption of glucose, and improves insulin sensitivity by increasing peripheral glucose uptake and utilization in skeletal muscles.

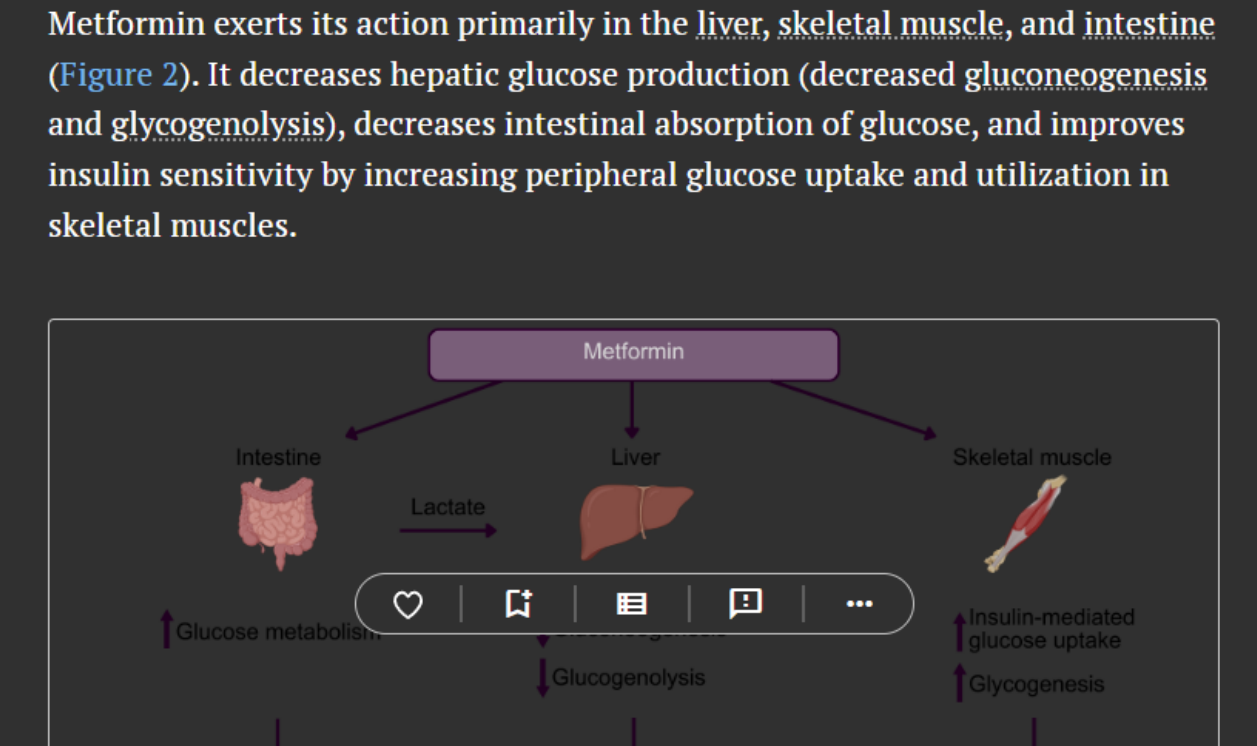

Figure 2

The mechanism of metformin in the liver and intestine is not completely understood. However, in skeletal muscle cells:

- Cellular glucose uptake and utilization is enhanced by stimulating AMPK.
  - AMPK is the protein in cells responsible for monitoring ratios of AMP to ATP and ADP to ATP.
- This then activates GLUT-4 translocation to the cell membrane.
  - GLUT-4 is the protein that moves glucose into muscle and fat cells in response to insulin.
- This activation of GLUT-4 by metformin leads to *increased glucose uptake by the cells* and decreased lipogenesis.

Advantages include weight loss, decreased triglyceride and low-density lipoprotein (LDL) cholesterol levels, lower blood pressure, and low risk for causing hypoglycemia. Metformin improves cardiovascular risk in obese patients with type 2 DM.

GLP-1 RA

Glucagon-like peptid mimic the effect of endogenous GLP-1, an incretin hormone released from the intestine after a meal. Examples include semaglutide, liraglutide, dulaglutide, and exenatide. These medications bind to GLP-1 receptors and help lower glucose through several complementary effects:

- Increased glucose-dependent insulin secretion from pancreatic β cells
- Decreased glucagon secretion from pancreatic α cells
- Slowed gastric emptying, which blunts postprandial glucose spikes
- Increased satiety, which can decrease food intake and promote weight loss

The phrase "glucose-dependent" is important. GLP-1 RAs stimulate insulin release most strongly when glucose levels are elevated, which is why they have a low risk of hypoglycemia when used without insulin or sulfonylureas. They are also useful in many patients with Type 2 DM and obesity because they can promote weight loss. Some GLP-1 RAs have demonstrated cardiovascular benefit, and current standards support GLP-1 RA use in selected patients with Type 2 DM and cardiovascular or kidney disease risk.

SGLT2 inhibitor

Sodium-glucose cotransporter 2 inhibitors, or SGLT2 inhibitors, lower blood glucose by increasing renal excretion of glucose. Examples include empagliflozin, dapagliflozin, and canagliflozin.

Under normal conditions, filtered glucose is largely reabsorbed in the proximal tubule through SGLT2 transporters. SGLT2 inhibitors block this transporter, so less glucose is reabsorbed and more glucose is excreted in the urine. This lowers serum glucose through an insulin-independent mechanism. Because their effect does not require pancreatic β-cell insulin release, SGLT2 inhibitors have a low risk of hyp or insulin secretagogues.

SGLT2 inhibitors can also cause modest weight loss because glucose calories are lost in the urine. They are especially important clinically because selected agents reduce heart failure hospitalization and slow chronic kidney disease progression in appropriate patients with Type 2 DM. ADA guidance recommends SGLT2 inhibitors for many patients with Type 2 DM and chronic kidney disease, including those with eGFR ≥20 mL/min/1.73 m², because they slow CKD progression and reduce heart failure risk.

CLINICAL CORRELATION

Not all GLP-1 receptor agonists and SGLT2 inhibitors have the same outcomes data, so avoid assuming benefit applies equally to every drug in the class. Among GLP-1–based therapies, **liraglutide, semaglutide, and dulaglutide** have demonstrated cardiovascular benefit, with **semaglutide** having especially strong evidence for kidney protection in patients with Type 2 DM and CKD.

Among SGLT2 inhibitors, **empagliflozin, dapagliflozin, and canagliflozin** have demonstrated important cardiovascular and kidney benefits, with SGLT2 inhibitors being especially high-yield for heart failure and CKD risk reduction.

What are the adverse effects of metformin, GLP-1 RA, and SGLT2 inhibitors? (LO#2)

Metformin, GLP-1 RAs, and SGLT2 inhibitors generally have a low risk of hypoglycemia when u y force insulin release regardless of glucose level. This distinguishes them from insulin and insulin secretagogues, such as sulfonylureas and meglitinides, which can cause hypoglycemia and weight gain. However, each medication class has important adverse effects.

Metformin

The most common adverse effects are:

- Gastrointestinal (GI) upset, nausea, flatulence, dyspepsia, abdominal pain, and/or diarrhea
  - These gastrointestinal effects are common for a patient just starting on metformin. However, they can be mitigated.
  - Most of these effects will self-resolve within 1-2 weeks, however, *starting at the lowest dose* and *slowly titrating* every 1-2 weeks can eliminate these effects.
  - *Reducing the dose*, if the patient is taking a mid- or high-range dose.
  - Switching to an *extended-release formulation* or taking it with food can mitigate these effects too.
- Metallic taste
  - Metformin is secreted in the saliva and can cause a metallic taste when a patient starts, or restarts the therapy.
  - This will self-resolve within a few weeks of consistent use.

CLINICAL CORRELATION

Does metformin help patients lose weight? This drug is usually associated with small weight loss, or ointestinal adverse effects can cause pa wear off the increased insulin sensitivity can help patients maintain, or have a neutral effect, on their weight.

How can the gastrointestinal side effects of metformin be mitigated?

Long-term use of metformin has been associated with reversible **vitamin B_{12} deficiency** and subsequent anemia and neuropathy. This can occur because metformin interferes with the absorption of vitamin B_{12} in the terminal ileum. The incidence rate of vitamin B_{12} deficiency in patients on metformin is about 7%, and it can be mitigated with vitamin B_{12} supplementation.

Like many medications, metformin has to be dose adjusted based on renal function. Specific recommendations include:

- eGFR of 30-45 m
  - Continuation may be appropriate at a reduced dose if the patient is already tolerating metformin.
  - Initiation of metformin is NOT recommended
- eGFR of < 30 mL/min/m2 is a *contraindication* to metformin therapy

These recommendations are in place because the risk of lactic acidosis greatly increases, which will be discussed in the next section.

How can metformin cause lactic acidosis? (LO#3)

The more serious side effect is lactic acidosis, which, although rare, can be fatal. In the intestine, metformin causes inhibition of mitochondrial electron transport. This effect then causes the cells in that area to favor *anaerobic glucose metabolism*, which causes an accumulation of lactate. The lactate production is also demonstrated in Figure 2 above. In most patients, i.e. those with normal renal function, the lactate is cleared without any issue. Because patients with decreased renal function are not able to clear lactate well, it predisposes them to lactic acidosis and has the restrictions/contraindications mentioned above.

The occurrence of lactic acidosis in relation to metformin use is also dose dependent, meaning as dose increases, the rate of lactic acidosis increases. Additionally, other disease states, like severe congestive heart failure or progressive liver disease, where large amounts of lactate are produced can have an accumulation effect with metformin and increase the risk of lactic acidosis. Being mindful of a patient's comorbidities and how they interrelate to their medications is always good practice.

What factors increase the risk of lactic acidosis with metformin use?

GLP-1 RA

The most common adverse effects of GLP-1 RAs are gastrointestinal, including nausea, vomiting, diarrhea, constipation, early satiety, and abdominal discomfort. These effects are usually most noticeable when therapy is started or when the dose is increased. Strategies to minimize these effects include starting at a low dose, increasing the dose gradually, eating smaller meals, stopping when full, and avoiding large high-fat meals that may worsen nausea.

A rare but important adverse effect is acute pancreatitis. Patients should be counseled to seek care for severe, persistent abdominal pain, especially if it radiates to the back or is associated with vomiting. GLP-1 RAs rarely cause hypoglycemia when used alone, but hypoglycemia risk can increase if they are combined with insulin or sulfonylureas.

CLINICAL CORRELATION

Do GLP-1 receptor agonists cause hypoglycemia? GLP-1 receptor agonists have a low risk of hypoglycemia when used alone because they increase insulin secretion in a glucose-dependent manner. This means they stimulate insulin release more strongly when blood glucose is elevated and have less effect when blood glucose is low.

However, hypoglycemia risk can increase when GLP-1 receptor agonists are combined with medications that directly lower glucose independent of glucose level, such as insulin or sulfonylureas. In those situations, the dose of insulin or the sulfonylurea may need to be reduced to prevent hypoglycemia.

SGLT1 Inhibitor

SGLT2 inhibitors increase the amount of glucose in the urine. This mechanism explains many of their adverse effects. Glucosuria can increase the risk of genital mycotic infections, such as vulvovaginal candidiasis or candidal balanitis. Patients can reduce this risk with good genital hygiene and by seeking treatment early if symptoms such as itching, irritation, discharge, or discomfort occur.

SGLT2 inhibitors can also cause osmotic diuresis, leading to increased urination, volume depletion, dizziness, or dehydration. Patients should be counseled to maintain adequate fluid intake and to be cautious during acute illness, poor oral intake, vomiting, diarrhea, or prolonged fasting.

A rare but serious adverse effect is euglycemic diabetic ketoacidosis. In this condition, ketoacidos extremely elevated. Risk is higher during prolonged fasting, acute illness, dehydration, major surgery, very low-carbohydrate intake, or states of low insulin availability. Patients should be counseled to hold the medication and seek medical guidance during significant acute illness or before major procedures, based on local protocols.

Table 2 below compares the adverse effects and important notes between these three antidiabetic medication classes.

Table 2

	Metformin	GLP-1 RA	SGLT2 inhibitor
Hypoglycemia risk (when used alone)	Low	Low	Low
Weight effect	Weight neutral or modest weight loss	Weight loss	Modest weight loss
Benefits	Low hypoglycemia risk, inexpensive, long clinical experience	Weight loss; cardiovascular benefit with selected agents	Heart failure and kidney protection with selected agents; cardiovascular benefit in appropriate patients
Common adverse effects	GI upset, diarrhea, nausea, metallic taste	Nausea, vomiting, diarrhea, constipation, abdominal discomfort	Genital mycotic infections, increased urination, volume depletion
Serious adverse effects	Lactic acidosis in high-risk patients; vitamin B12 deficiency with long-term use	Rare pancreatitis; hypoglycemia risk increases if combined with insulin or sulfonylureas	Euglycemic DKA; dehydration/volume depletion
Strategies to minimize adverse effects	Start low, titrate slowly, take with food, use extended-release formulation, monitor renal function and B12 when appropriate	Start low, titrate slowly, eat smaller meals, avoid large high-fat meals, counsel on pancreatitis symptoms	Counsel on hydration and genital hygiene; hold during significant acute illness, prolonged fasting, or peri-procedural periods per protocol

BACK TO INTRODUCTION ↑

CASE CONNECTION

Returning to TK. It is frustrating for many patients to be adherent to their medication regimen but experience bothersome adverse effects. You would want to acknowledge his concerns, encourage him for taking the medication consistently, and explain that his symptoms are common when starting metformin.

You could offer several strategies to reduce the gastrointestinal adverse effects while maintaining therapy: lower the dose, take metformin with food, titrate more slowly, or switch to an extended-release formulation. You could also explain that these effects often improve within the first few weeks of consistent use.

As TK's diabetes care continues, it is also helpful to explain that metformin is only one medication option for Type 2 DM. Other commonly used classes include GLP-1 receptor agonists and SGLT2 inhibitors. These medications lower glucose through different mechanisms and may be especially useful in patients with obesity, cardiovascular disease, heart failure, or chronic kidney disease. The best medication choice depends on glycemic goals, comorbidities, kidney function, adverse effect risk, cost, access, and patient preferences.

Summary

- Type 2 Diabetes Mellitus is characterized by insulin resistance, high glucagon levels, and progressive insufficiency of pancreatic insulin secretion.
- Metformin, GLP-1 receptor agonists, and SGLT2 inhibitors are common non-insulin medications used in Type 2 DM, but they are not the only available antidiabetic medications.
- Metformin is a biguanide that increases insulin sensitivity, decreases hepatic glucose production, decreases intestinal glucose absorption, and improves peripheral glucose uptake.
- GLP-1 receptor agonists mimic incretin signaling, increasing glucose-dependent insulin secretion, decreasing glucagon secretion, slowing gastric emptying, and increasing satiety.
- SGLT2 inhibitors block glucose reabsorption in the proximal tubule, increasing urinary glucose excretion and lowering serum glucose through an insulin-independent mechanism.
- Metformin commonly causes gastrointestinal upset and metallic taste; these effects can be minimized by starting at a low dose, titrating slowly, taking with food, or switching to an extended-release formulation.
- Long-term metformin use may cause reversible vitamin B12 deficiency, which may require monitoring and supplementation.
- Lactic acidosis is a rare but serious adverse effect of metformin, with increased risk in patients with high doses, significantly decreased renal function, severe heart failure, or progressive liver disease.
- GLP-1 receptor agonists commonly cause gastrointestinal adverse effects and rarely pancreatitis; starting low and titrating slowly can improve tolerability.
- SGLT2 inhibitors can cause genital mycotic infections, volume depletion, and rare euglyce should emphasize hydration, hygiene, and holding therapy during significant acute illness or fasting states.
- GLP-1 receptor agonists and SGLT2 inhibitors are especially important because selected agents provide cardiovascular and kidney benefits in appropriate patients with Type 2 DM.

Review Questions

1. Which of the following best matches each medication class with its primary glucose-lowering mechanism?

A. ○ Metformin increases renal glucose excretion; GLP-1 RAs increase insulin sensitivity; SGLT2 inhibitors decrease gastric emptying
B. ○ Metformin increases insulin sensitivity; GLP-1 RAs increase glucose-dependent insulin secretion; SGLT2 inhibitors increase renal glucose excretion
C. ○ Metformin directly stimulates pancreatic insulin release; GLP-1 RAs block proximal tubular glucose reabsorption; SGLT2 inhibitors decrease hepatic gluconeogenesis
D. ○ Metformin decreases glucagon secretion; GLP-1 RAs increase urinary glucose excretion; SGLT2 inhibitors stimulate GLUT-4 translocation
E. ○ Metformin blocks carbohydrate absorption; GLP-1 RAs inhibit DPP-4; SGLT2 inhibitors stimulate pancreatic β cells

Explanation (requires correct answer)

2. A 56-year-old patient with Type 2 DM and obesity is taking metformin but remains above their A1c goal. The clinician wants to add a medication that can improve glyce d has low risk of hypoglycemia when not combined with insulin or sulfonylureas. Which mechanism best describes this add-on therapy?

A. ○ Closing ATP-sensitive potassium channels on pancreatic β cells
B. ○ Blocking SGLT2 in the proximal tubule
C. ○ Activating GLP-1 receptors to increase glucose-dependent insulin secretion and satiety
D. ○ Directly replacing endogenous insulin
E. ○ Inhibiting α-glucosidase enzymes in the intestinal brush border

Explanation (requires correct answer)

3. A patient with Type 2 DM and chronic kidney disease is started on a medication that lowers glucose by increasing glucosuria. Which adverse effect should the patient be counseled about based on this mechanism?

A. ○ Severe hypoglycemia due to glucose-independent insulin release
B. ○ Genital mycotic infections and volume depletion
C. ○ Fluid retention and worsening heart failure
D. ○ Vitamin B12 deficiency and metallic taste
E. ○ Cough an
E. ○ Cough and angioedema

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Explain the mechanism of action of metformin, GLP-1 receptor agonists, and SGLT2 inhibitors in the context of managing Type 2 Diabetes.
02 ☐ Discuss the adverse effects of metformin, GLP-1 receptor agonists, and SGLT2 inhibitors and discuss strategies to minimize/prevent such effects.
03 ☐ Describe the potential risk of developing lactic acidosis based on metformin pharmacokinetics.

Objective Confidence: 0/3